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Escherichia coli Mutant Strain with Altered Expression of the Tryptophan Operon: Isolation and Preliminary Characterization

PETER H. POUWELS* AND HARRY J. SCHOLTEN
Medical Biological Laboratory TNO, 2280 AA Rijswijk Z.H., The Netherlands

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From a strain of *Escherichia coli* with two copies of the tryptophan (*trp*) operon and one copy of the lactose (*lac*) operon, under control of one of the *trp* regulatory elements, we have isolated a mutant which does not grow in a medium containing 19 amino acids, unless tryptophan is added, and which cannot ferment lactose. The apparent pleiotropic nature of the mutation(s) is indicated by the very slow growth of mutant bacteria on minimal-medium agar supplemented with glucose and tryptophan. The amount of the *trp* enzymes (anthranilate synthetase and tryptophan synthetase) and *trp* messenger ribonucleic acid is reduced severalfold in the mutant compared to the isogenic wild-type strain, whereas the enzymes tryptophanyl-transfer ribonucleic acid synthetase and glucose 6-phosphate dehydrogenase remain the same. The incorporation of radioactive label into pulse-labeled but not into stable ribonucleic acid is significantly lower. Our results suggest that in the mutant organism the control of transcription of some operons, including the *trp* operon, is modified. An alternative explanation is that mutant bacteria contain a ribonuclease with increased activity for some messenger ribonucleic acid species.

Control of transcription in *Escherichia coli* is exerted mainly at two levels: initiation and termination of RNA synthesis (2, 6). Regulation of RNA synthesis by varying the frequency of initiation of transcription is controlled negatively by the repressor (8, 19) and controlled positively by control factors such as the cyclic AMP receptor protein (5). Control of RNA synthesis by termination of transcription takes place at a site located at the promotor distal end of the regulatory region of the operon (3) and involves the transcription termination factor Rho (16).

To gain insight into the nature of these two controls and the mechanism of action of the cellular constituents that control transcription, it seems advantageous to isolate mutant organisms with altered control of transcription. In this paper we describe the isolation and preliminary characterization of such a mutant. In the accompanying paper we report on studies of RNA synthesis in vitro with partially purified RNA polymerase prepared from the mutant organism.

MATERIALS AND METHODS

Bacteria and bacteriophages. All experiments were carried out with derivatives of *E. coli* K-12 *Δlac174 Δ(trp-tonB-Φ80dlac)205 trpR rpsL* F'*trp* (2). The *tonB* deletion in this strain provides for a *trp-lac* fusion (W205) which effectively puts a transposed *lac* operon under *trp* control. The fusion is to the distal part of the *trp* operon, which in this strain remains functional (12). The strain is thus diploid for *trp*. The strain will be indicated as TL105 Trp$^+$ Lac$^+$. From this strain a mutant was isolated (TL105 Trp$^-$ Lac$^-$) which requires tryptophan for growth and does not ferment lactose at an appreciable rate. The mutant was obtained after mutagenic treatment with ethyl methane sulfonate and selection for bacteria which survived penicillin treatment in a synthetic medium containing 19 amino acids (a 0.25% solution of Casamino Acids) but lacking tryptophan. Phages λpolEDCBA-6 and λEDCBA-BG2 are tryptophan-transducing strains of phage λ (13). Phages λpolEDCBA-6 contains the entire *trp* operon, including the regulatory region (pol), and λEDCBA-BG2 contains the entire *trp* operon, except for the promotor proximal region (~70 nucleotides; 18).

Enzyme assays. Anthranilate synthetase (product of genes *trpE* and *trpD*) was measured according to Ito et al. (9). Tryptophan synthetase (product of genes *trpB* and *trpA*) was measured according to Smith and Yanofsky (17). Enzyme activities are expressed as units of activity per milligram of protein. One unit of each of the *trp* enzymes is defined as the amount of enzyme which catalyzes the conversion of 0.1 μmol of substrate to product in 20 min at 37°C under the conditions of the experiment. Protein concentrations were determined by the method of Lowry et al. (11).

Preparation of ^{3}H-labeled RNA in vivo. Bacteria were cultivated at 37°C in Vogel-Bonner (21) salt medium (5 ml) supplemented with 0.5% glucose, 0.25% Casamino Acids, 1 μg of thiamine per ml, and 50 μg of tryptophan per ml. When the bacterial culture had reached a density of 1×10^8 to 2×10^8, RNA was pulse-labeled for 60 s with [^{3}H]uracil (50 μCi/ml; 53 Ci/mmol). RNA was isolated as described by Summers (20). After precipitation with alcohol, [^{3}H]RNA was dissolved in 2.0 ml of 2× SSC (0.3 M NaCl, 0.03 M sodium citrate) plus 10 mM Tris (pH 7.6), heated for 3 min at 100°C, quickly cooled in ice, and filtered three times through nitrocellulose filters (Schleicher and Schuell, BA85) to reduce nonspecific binding to filters in the hybridization assay.

Determination of the amount of total *trp* mRNA, *trpE-A* mRNA, and *trp* pol mRNA. The amount of total *trp* mRNA (encompassing both regulatory and structural genes) was determined by hybridization of [^{3}H]RNA to excess of *l*-λpolEDCBA-6 DNA, as described before (14). The amount of *trpE-A* mRNA was determined by hybridization of [^{3}H]RNA to excess *l*-λEDCBA-BG2 DNA. The value for *trp* pol mRNA (the regulatory region) was obtained by subtracting *trpE-A* mRNA from total *trp* mRNA.

The incorporation of [^{3}H]uracil into RNA. To determine the incorporation of [^{3}H]uracil into stable RNA, bacteria were cultivated at 37°C in Vogel-Bonner minimal medium (5 ml) supplemented with 0.1% Casamino Acids, 100 μg of tryptophan per ml, and 25 μCi of [^{3}H]uracil (50 μg). At various periods of time after addition of the radioisotope, samples (100 μl) were taken and the amount of radioactivity incorporated into acid-precipitable material was determined. To pulse-label bacteria, [^{3}H]uracil (2 μCi; 2.4 μg) was added, and the amount of [^{3}H]uracil incorporated into acid-precipitable material during a 30-s period was measured.

RESULTS

Isolation of a strain of *E. coli* with altered control of transcription of the *trp* genes. A strain of *E. coli* with altered control of transcription of the *trp* genes was isolated, after mutagenic treatment, by selecting for bacteria which require tryptophan for growth and do not ferment lactose. To improve the selection and to select against tryptophan-requiring bacteria, as such, due to a mutation in the *trp* operon itself, we used a strain which was diploid for the *trp* operon. As a second improvement, the lactose (*lac*) genes had been transposed and brought under control of the *trp* regulatory elements. Mutants that are both Lac$^-$ and Trp$^-$ require tryptophan not because of a deficiency either in the charging of tRNA^{Trp} or in the transfer of tryptophan from tRNA to protein, but because of some other deficiency.

One such strain was obtained. In rich medium, e.g., L-broth or synthetic medium containing all 20 amino acids, the doubling time of the mutant strain was 20 to 50% longer than that of the parent strain. Mutant bacteria were unable to grow in synthetic media containing 19 amino acids, but lacking tryptophan. On Vogel-Bonner medium agar supplemented with glucose and tryptophan, the bacteria grew very slowly, and small colonies appeared only after incubation for 5 to 7 days at 37°C. The mutation that renders the bacteria Trp$^-$ is not conditionally lethal, and in liquid media the bacterial cultures only attained low densities, suggesting that the mutant strain requires another metabolite besides tryptophan.

Activity of *trp* enzymes in bacterial extracts. The activity of anthranilate synthetase, encoded by genes *trpE* and *trpD*, and that of tryptophan synthetase, encoded by *trpB* and *trpA*, was assayed in extracts prepared from mutant and wild-type bacteria which were in the exponential phase of growth. The results (Table 1) show that mutant bacteria contain significant but considerably reduced amounts of the two enzymes. A similar reduction in specific activity of both was found, indicating that in mutant bacteria, as in wild-type bacteria, the enzymes of the *trp* operon are coordinately expressed.

These results suggest that the phenotype of the bacteria is not caused by a mutation in *trpE, D,* or *B*. A strain with a deletion of the entire *trp* operon and the adjacent gene *tonB* can be transduced to tryptophan prototrophy with a P1 lysate grown on TL105 Trp$^-$ Lac$^-$ (results not shown), indicating that the *trp* operon of the mutant strain is not mutated. In addition, both mutant and wild-type bacteria contained a *trpR* mutation (repressor negative), ruling out an effect of the repressor on the rate of synthesis of the *trp* enzymes in the mutant bacteria.

To determine whether the reduced activity of mutant bacteria is specific for the *trp* operon, we also assayed the activity of two other enzymes. First, we assayed the activity of tryptophanyl-tRNA synthetase. Bacteria that have low levels of this enzyme are phenotypically Trp$^-$ (7, 10). Second, we assayed the activity of glucose 6-

TABLE 1. *Specific activity of tryptophan enzymes in extracts from mutant and wild-type bacteria*[a]

Bacteria	ASase	TSase
Wild type	0.7	8.7
Mutant	0.2	2.1

[a] Anthranilate synthetase (ASase) and tryptophan synthetase (TSase) were assayed in extracts prepared from mutant or wild-type bacteria as described in the text. The bacteria were cultivated in L-broth and were harvested in the exponential phase of growth. Data are expressed as units of activity (amount of enzyme that catalyzes the conversion of 0.1 μmol of substrate to product in 20 min at 37°C) per milligram of protein.

phosphate dehydrogenase, which is not related to tryptophan biosynthesis and which is synthesized constitutively in *E. coli.* The activities of both enzymes were not significantly different in extracts from mutant and wild-type bacteria (not shown). The results therefore indicate that the mutation specifically affects the synthesis of enzymes of the *trp* operon. They also indicate that the requirement for tryptophan of the bacteria is not caused by a reduced rate of protein synthesis due to a lack of tryptophanyl-tRNA synthetase.

The synthesis of *trp* mRNA in mutant and wild-type bacteria. To ascertain whether the reduced capacity of mutant bacteria to synthesize *trp* enzymes is due to a deficiency to synthesize *trp* mRNA or to a reduced efficiency to translate this RNA, we studied the formation of *trp* mRNA in mutant and wild-type bacteria. The amount of radioactivity incorporated into ^{3}H-labeled *trp* mRNA was determined by hybridization of [^{3}H]RNA to the *l*-DNA strand from phage λpolEDCBA-6, which contains the entire *trp* operon. The results (Table 2) show that this amount is four- to fivefold lower in mutant than in wild-type bacteria. The four- to fivefold lower incorporation reflects a reduction of the net synthesis of ^{3}H-labeled *trp* mRNA, since the fraction of [^{3}H]RNA that is *trp* mRNA is reduced to a similar extent. The reduced amount of ^{3}H-labeled *trp* mRNA of mutant bacteria could be due to a smaller rate of synthesis or a greater rate of degradation of *trp* mRNA, or both. Assuming that it is the rate of synthesis that is altered in mutant bacteria (see also accompanying paper [15]), we wished to determine whether this difference is caused by a reduced rate of initiation of transcription of the *trp* genes or by a more efficient termination of transcription at the termination site located at the promotor distal end of the regulatory region of the operon (3).

TABLE 2. *Amount of trp mRNA in mutant and wild-type bacteria*[a]

Bacteria	*trp* mRNA (disintegrations per min)	Percent total [^{3}H]RNA
Wild type	6,413	1.0
Mutant	1,544	0.2

[a] The amount of *trp* mRNA was assayed in mutant and wild-type bacteria by hybridization of [^{3}H]RNA, synthesized in vivo, and excess *l*-DNA from λpol-EDCBA-6. Bacteria were cultivated in Vogel-Bonner minimal medium (5 ml) supplemented with 0.5% glucose, 0.25% Casamino Acids, 1 μg of thiamine per ml, and 50 μg of tryptophan per ml, and equivalent amounts of mutant and wild-type bacteria were pulse-labeled with [^{3}H]uracil (50 Ci/ml) for 60 s at a density of 2×10^8 to 3×10^8/ml. The results have been corrected for nonspecific hybridization of [^{3}H]RNA to excess *l*-λDNA.

To this end we determined, by RNA-DNA hybridization, the ratio of the amounts of *trpE-A* mRNA and *trp* pol mRNA synthesized. This ratio reflects the efficiency with which termination at the transcriptional termination site occurs. The amount of *trpE-A* mRNA was determined by hybridization of [^{3}H]RNA to *l*-λEDCBA-BG2 DNA, which contains the entire *trp* operon except for the promotor proximal end (~70 nucleotides). The amount of *trp* pol mRNA can then be computed from the difference of the hybridization data with *l*-λpolEDCBA-6 DNA and *l*-λEDCBA-BG2 DNA. The results (not shown) indicate that the ratios of *trpE-A* mRNA to *trp* pol mRNA do not differ significantly for the two strains, suggesting that the reduced rate of synthesis of total *trp* mRNA cannot be explained by an increase of the efficiency of termination in mutant cells.

The rate of incorporation of [^{3}H]uracil into RNA in mutant and wild-type bacteria. The results presented in the previous section indicate that in mutant bacteria the synthesis of at least one species of RNA, viz. total *trp* mRNA, is impaired. To investigate whether the synthesis of other species of RNA is also affected by the mutation, we measured the incorporation of [^{3}H]uracil into stable RNA, which predominantly consists of rRNA, and into pulse-labeled RNA, comprising mRNA and stable RNA (Fig. 1). To facilitate the interpretation the results in Fig. 1A (incorporation of [^{3}H]uracil into stable RNA) have been expressed as the ratio of the rates of incorporation of [^{3}H]uracil into stable RNA versus the absorbancy at 700 nm (Table 3). Similarly, the results in Fig. 1B have been expressed as the ratio of the incorporation of [^{3}H]uracil into pulse-labeled RNA versus the absorbancy at 700 nm (Table 3).

Our results show that early in the exponential phase of growth the incorporation of [^{3}H]uracil into stable RNA and pulse-labeled RNA is comparable in mutant and wild-type bacteria. However, later in the exponential phase of growth the rate of incorporation of [^{3}H]uracil into pulse-labeled RNA was considerably smaller in mutant than in wild-type bacteria. At this stage of growth only a small difference was found for the rates of incorporation of [^{3}H]uracil into stable RNA.

DISCUSSION

In this paper we have described a novel selection technique for the isolation of bacteria with altered control of gene expression. The tech-

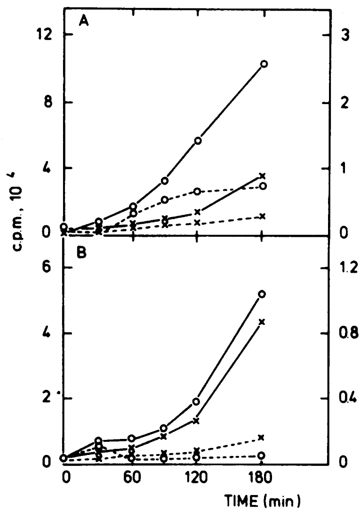

FIG. 1. *The incorporation of [³H]uracil into stable RNA (A) and pulse-labeled RNA (B) in mutant and wild-type bacteria was measured as described in the text. The increase of the absorbancy at 700 nm (A_{700}) was determined spectrophotometrically. (A) Counts per minute (c.p.m.) with time after addition of [³H]-uracil at $t = 0$; (B) counts of [³H]uracil incorporated in 30-s pulses at the times indicated. Symbols:* (×——×) *wild type,* A_{700}; (○——○) *wild type, counts per minute;* (×-----×) *mutant,* A_{700}; (○-----○) *mutant, counts per minute.*

TABLE 3. *Rate of incorporation of [³H]uracil into stable and pulse-labeled RNA in mutant and wild-type bacteria*[a]

Incubation time (min)	Stable RNA (Δcpm × 10^3/ΔA_{700})[b]		Pulse-labeled RNA (cpm × 10^3/ΔA_{700})	
	Wild type	Mutant	Wild type	Mutant
60	165	157	150	86
120	190	129	83	31
180	124	128	62	22

[a] The data from Fig. 1 are expressed as the ratios of the rate of incorporation of [³H]uracil into RNA (counts per minute [cpm]) versus that of absorbancy at 700 nm (A_{700}).

[b] Δ values represent the values measured at the indicated times minus the value at $t = 0$ min.

nique has been used for the isolation of a strain of *E. coli* with altered expression of the *trp* operon. The selection is based on the simultaneous shutoff of two functions that are controlled by a single promotor. In our case this was achieved by using a strain in which the *lac* operon was brought under the control of the *trp* promotor. In combination with procedures such as that described by Casadeban (4) for the transposition and fusion of *lac* genes to selected genes, this method might be valuable for the isolation of mutants with altered control elements governing those genes.

The mutant strain described in this paper was isolated as a tryptophan-requiring strain by selecting for bacteria which did not grow in a medium containing a mixture of all 19 amino acids, but lacking tryptophan. Measurement of the activity of the *trp* biosynthetic enzymes in the mutant strain revealed a four- to fivefold reduction compared with that of wild-type bacteria, but even this reduced activity seems too high to explain the Trp⁻ phenotype of the bacteria. Presumably the mutation that reduces the rate of synthesis of the *trp* enzymes also directly, or indirectly, impairs the utilization of tryptophan for the synthesis of proteins. The pleiotropic character of the mutation(s) is further indicated by the reduced growth rate of the bacteria in rich media containing tryptophan and by the reduced rate of incorporation of [³H]uracil into pulse-labeled RNA. To better understand the pleiotropic character, however, a thorough genetic characterization of the mutant strain is required.

Results of experiments presented in this paper suggest that the mutation reduces the rate of accumulation of total *trp* mRNA. It was shown also that the rate of incorporation of [³H]uracil into pulse-labeled RNA but not into stable RNA is impaired. It appears that a transcriptional control mechanism is modified in the mutant strain, either by a mutation affecting the activity of RNA polymerase or by modification of a factor which controls transcription. An alternative explanation would be that the mutant bacteria contain an RNase with increased activity for some mRNA species.

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LITERATURE CITED

1. **Adhya, S., and M. Gottesman.** 1978. Control of transcription termination. Annu. Rev. Biochem. **47**:967–996.
2. **Barnes, W. M., R. B. Siegel, and W. S. Reznikoff.** 1974. The construction of λ transducing phages containing deletions defining regulatory elements of the *lac* and *trp* operons in *E. coli*. Mol. Gen. Genet. **129**:201–215.
3. **Bertrand, K., C. Squires, and C. Yanofsky.** 1976. Transcription termination in vivo in the leader region of the tryptophan operon of *Escherichia coli*. J. Mol. Biol. **103**:319–337.
4. **Casadeban, M. J.** 1976. Transposition and fusion of the lac genes to selected promotors in Escherichia coli using bacteriophage lambda and mu. J. Mol. Biol. **104**:541–555.
5. **De Crombrugghe, B., B. Chen, W. Anderson, M. Gottesman, R. Perlman, and I. Pastan.** 1971. Role

of cyclic adenosine 3',5'-monophosphate and the cyclic adenosine 3',5'-monophosphate receptor protein in the initiation of lac transcription. J. Biol. Chem. **246**:7343-7348.

6. **Dickson, R. C., J. Abelson, W. M. Barnes, and W. S. Reznikoff.** 1975. Genetic regulation: the lac control region. Science **187**:27-35.
7. **Doolittle, W. F., and C. Yanofsky.** 1968. Mutants of *Escherichia coli* with an altered tryptophanyl-transfer ribonucleic acid synthetase. J. Bacteriol. **95**:1283-1294.
8. **Gilbert, W., and B. Müller-Hill.** 1967. The lac operator in DNA. Proc. Natl. Acad. Sci. U.S.A. **58**:2415-2421.
9. **Ito, J., E. Cox, and C. Yanofsky.** 1969. Anthranilate synthetase, an enzyme specified by the tryptophan operon of *Escherichia coli*: purification and characterization of component I. J. Bacteriol. **97**:725-733.
10. **Kano, Y., A. Matsushiro, and Y. Shimura.** 1968. Isolation of the novel regulatory mutants of the tryptophan biosynthetic system in *Escherichia coli*. Mol. Gen. Genet. **102**:15-26.
11. **Lowry, O. H., N. J. Rosebrough, A. L. Farr, and R. J. Randall.** 1951. Protein measurement with the Folin phenol reagent. J. Biol. Chem. **193**:265-275.
12. **Mitchell, D. H., W. S. Reznikoff, and J. R. Beckwith.** 1975. Genetic fusions defining *trp* and *lac* operon regulatory elements. J. Mol. Biol. **93**:331-350.
13. **Pannekoek, H., W. J. Brammar, and P. H. Pouwels.** 1975. Punctuation of transcription in vitro of the tryptophan operon of *Escherichia coli*. A novel type of control of transcription. Mol. Gen. Genet. **136**:199-214.
14. **Pannekoek, H., B. Perbal, and P. H. Pouwels.** 1974. The specificity of transcription in vitro of the tryptophan operon of *Escherichia coli*. II. The effect of Rho factor. Mol. Gen. Genet. **132**:291-306.
15. **Pouwels, P. H., and G. de Groot.** 1979. *Escherichia coli* mutant strain with altered expression of the tryptophan operon: ribonucleic acid synthesis in vitro. J. Bacteriol. **139**:398-403.
16. **Roberts, J. W.** 1969. Termination factor for RNA synthesis. Nature (London) **224**:1168-1171.
17. **Smith, O., and C. Yanofsky.** 1962. Enzymes involved in biosynthesis of tryptophan. Methods Enzymol. **5**:801-806.
18. **Squires, C., F. Lee, K. Bertrand, C. L. Squires, M. J. Bronson, and C. Yanofsky.** 1976. Nucleotide sequence of the 5' end of Tryptophan messenger RNA of *Escherichia coli*. J. Mol. Biol. **103**:351-358.
19. **Squires, C. L., F. D. Lee, and C. Yanofsky.** 1975. Interaction of the *trp* repressor and RNA polymerase with the *trp* operon. J. Mol. Biol. **92**:93-111.
20. **Summers, W. C.** 1970. A simple method for extraction of RNA from *E. coli* utilizing diethyl pyrocarbonate. Anal. Biochem. **33**:459-463.
21. **Vogel, H., and D. M. Bonner.** 1956. Acetylornithinase of *Escherichia coli*: partial purification and some properties. J. Biol. Chem. **218**:97-106.